
STAIN NORMALIZATION IN HISTOPATHOLOGY USING GAN

DEEP LEARNING FOR VISUAL RECOGNITION

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ABSTRACT

We demonstrate that CycleGAN-based domain adaptation improves classification accuracy by 30.1 percentage points across unmatched domains¹. The data used is a subset from CAMELYON17. Quantitative metrics show successful domain adaptation, but we further investigate the interpretability of the classification model with Grad-CAM. We find that while the model ignores the background, it relies heavily on surrounding tissue instead of looking at the tumor tissue when classifying, highlighting the necessity of interpretable models.

Keywords Deep Learning · Generative Adversarial Networks · Grad-CAM · CNN · Stain Normalization

1 Introduction

Histopathology is the study of diseases by examining tissues under a microscope. Traditionally, this is done by a pathologist trained to classify diseases in tissues, but using computer-aided diagnosis (CAD) can be a helpful way to reduce the workload of experts. One common problem in the field is the difficulty of making a generalized algorithm that works on any kind of histopathological images. This is because these images show a lot of variance in color, quality, illumination etc. due to the differences in staining protocols (chemical protocols in which the structure of the tissue is amplified in color and clarity). Consequently, this variance hinders general approaches to image analysis, which is why image normalization techniques are needed. Our project focuses on such a technique.

As an example use case, consider hospital A that has trained a model and wants to provide it to hospital B . Hospital B might not have enough data to train their own model, but at the same time, their data will not work on the model created by hospital A due to the variance in the data. Using unpaired domain adaptation, we can adapt the data such that the model can still be used.

We aim to train a classifier that detects tumor cells given training data from a source domain (hospital A), then the classifier must also correctly classify data from the source domain, but also on data from a target domain (hospital B).

In our project, we use Cycle-Consistent Generative Adversarial Networks (CycleGAN) to accomplish this adaptation. CycleGAN is particularly well-suited for this task because it functions without paired labels, i.e., we don't need the ground truth when training our model, which makes data acquisition more feasible.

2 Related work

Studying methods to aid quantitative analysis of histopathological images is nothing new. Some studies use color matching based methods, where the overall color statistics of one image is matched to another, creating a color-color mapping between staining methods. Reinhard et al. [2001] does this by matching color brightness and color tone. This is a problem, as it ignores the underlying structure of the stain and the fact that different stains contribute differently to the final color.

¹The code is available at <https://gitlab.au.dk/> given access to the AU Gitlab instance.

Other methods use a stain-separation approach where normalization is done on each stain (i.e. H&E) separately. Macenko et al. [2009] does this by converting the RGB images into an Optical Density space, then uses SVD to estimate the stain vectors before normalizing them.

Vahadane et al. [2016] follow a similar idea but use sparse non-negative matrix factorization to achieve more stable stain separation.

More recently, deep learning based methods have become popular, especially after the introduction of GAN by Goodfellow et al. [2014] and CycleGAN by Zhu et al. [2017a] which made it possible to do stain normalization without using paired data. Shaban et al. [2018] proposed StainGAN which was a breakthrough at the time and since then similar models have shown very strong results.

For example Runz et al. [2021] used CycleGAN on data collected at the Institute of Pathology, Medical Faculty Mannheim, and managed to get high accuracy.

Shrivastava et al. [2019] proposed a Self-Attentive Adversarial Stain Normalization method, in which they found that adding self-attention layers to their GAN helped the model generate more vibrant results that preserved medically significant artifacts.

3 Methods

In this section, we describe how we did image-to-image translation between two domains of the CAMELYON17 dataset [DIAG 2017]. We begin by describing how the dataset was prepared for training, which includes preprocessing procedures, data splits, and augmentation strategies. Then we explain the CycleGAN architecture, how we used it, and the challenges we encountered along the way. We then describe the full domain adaptation pipeline in which we do tumor classification on generated data. Following this we show which experiments we conducted and how we managed to progress through the results of those experiments.

3.1 Dataset preparation

The dataset used in this project is a subset of the data from CAMELYON17 dataset [DIAG 2017] accessed via AWS [cam]. CAMELYON17 contains whole-slide images (WSIs), high-resolution digital scans, of lymph node sections collected from five different medical centers.

For this project, we restrict the dataset to WSIs from two centers (center_0 and center_2) and treat each as a separate domain. This is motivated by the fact that pathology centers differ in tissue preparation, staining protocols, and scanning equipment [Bejnordi 2017], which introduces significant visual domain shift.

Center_0 is treated as Domain A and center_2 as Domain B. As WSIs are extremely large (typically a few gigabytes), we extract 128×128 patches by sliding a 128×128 window across each slide with stride equal to the patch size, as illustrated in Figure 1.

Tumor slides are only processed if XML annotations are available, as they annotate the tumor regions, also called tumor polygons. Furthermore, patches containing less than 50% tissue (either normal or tumor) are discarded to avoid background-only samples and the tumor patches are discarded if they contain less than 2% tumor tissue. We made sure to extract the same amount of patches, which results in a dataset of 2070 tumor patches and 2070 normal patches per center. This was done so that any cross-domain performance gap is more plausibly due to stain differences, not data imbalance.

Each extracted patch is saved as a JPG file together with a JSON metadata file containing: the slide identifier, center and domain labels, class label, tumor fraction, and tumor polygons (when applicable). Figure 2 shows an example of tumor and normal patches from center_0 and center_2.

All data preparation steps are automated through the cam17_data_extraction module. The full process (download, filtering, patch extraction, metadata generation) takes approximately three to five hours, depending on network speed and disk performance. For convenience, a preprocessed dataset is also available as a compressed archive at <https://>

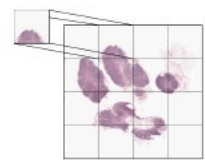


Figure 1: Patches from WSI.

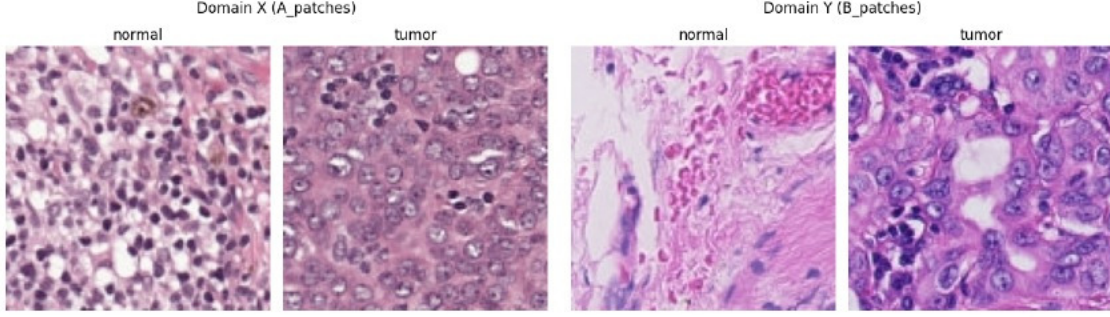


Figure 2: Patches from both domains showing an example of a tumorous and normal slide

3.2 CycleGAN

With the dataset prepared into source and target domains, we proceed to detail the CycleGAN architecture used to map between them. It was originally proposed by [Zhu et al. \[2017a\]](#), and is an adversarial approach to unpaired image-to-image translation. Unpaired in the sense that there is no feature-label pairs as in standard supervised learning.

As described in the previous section, the data we work with comes from different medical centers, resulting in varying coloring and scaling of images depending on the type of microscopes and staining processes used. However, the semantic components of the images are still the same, so the goal is to make classification on these semantic components ignoring the fact that images look different. Thus, CycleGAN will make translations of images from one domain to another in order to train a single classifier that works on both domains. We proceed with a brief explanation of how CycleGANs work.

The idea is to train a mapping $G : X \rightarrow Y$ such that $\hat{y} = G(x)$ is indistinguishable from images $y \in Y$ by an adversary D , trained to classify $\hat{y}$ apart from y . The hope is that this game between the generator G and discriminator D reaches an equilibrium in which the generated images $G(x)$ not only fool D but are essentially an empirical generalization of the true distribution Y . However, this equilibrium does not guarantee a meaningful translation, G might instead learn to produce only a small subset of possible outputs that consistently fool D , failing to capture the full diversity of the target distribution. This is a common pitfall of GANs referred to as *mode collapse*. [Zhu et al. \[2017a\]](#) introduced cycle consistency to constrain the mapping and prevent arbitrary permutations, which also helps mitigate mode collapse. The idea is to make inverse translations, i.e. if we have $G : X \rightarrow Y$ then we must also train an inverse model $F : Y \rightarrow X$. These are both trained simultaneously by adding a *cycle consistency loss* that encourages $F(G(x)) \approx x$ and $G(F(y)) \approx y$. This loss is combined with the adversarial losses as described in the original GAN paper [Goodfellow et al. \[2014\]](#). Figure 3 summarizes all components.

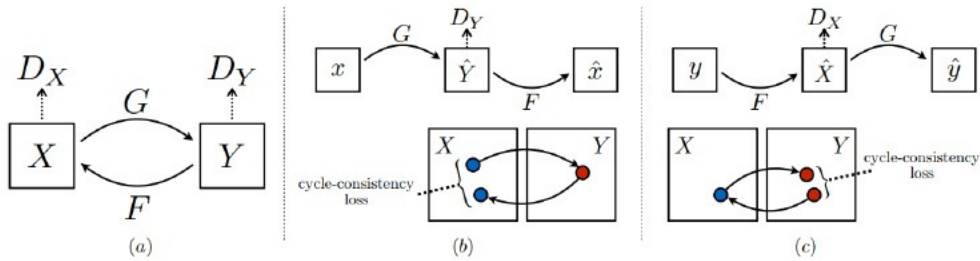


Figure 3: This figure from [Zhu et al. \[2017a\]](#) shows all the components of CycleGAN. (a) shows the two mapping functions (generators) G and F with their associated discriminators D_X and D_Y . The discriminators encourages the generators to translate their corresponding domain such that they are indistinguishable from the true domain. (b) shows a forward cycle consistency loss $F(G(x)) \approx x$, (c) shows a backward cycle consistency loss $G(F(y)) \approx y$.

Adversarial loss

We construct the objective function by first looking at the adversarial loss [Goodfellow et al. \[2014\]](#). For G and D_Y we express the loss as

$$\mathcal{L}_{\text{ADV}}(G, D_Y, X, Y) = \mathbb{E}_{y \sim \mathcal{D}(Y)} [\log D_Y(y)] + \mathbb{E}_{x \sim \mathcal{D}(X)} [\log(1 - D_Y(G(x)))]$$

where $D_Y(y) \in (0, 1)$ is high if D_Y believes y is real, so the first term in $\mathcal{L}_{ADV}$ is high if it correctly identifies real images, and the second term is high if D_Y correctly identifies fake images. Thus, the discriminator (our adversary) tries to maximize this loss while the generator tries to minimize it, i.e. $\min_G \max_{D_Y} \mathcal{L}_{ADV}(G, D_Y, X, Y)$. Similarly, we make a loss function for the generator F and its discriminator D_X , i.e. $\min_F \max_{D_X} \mathcal{L}_{ADV}(F, D_X, Y, X)$.

Cycle Consistency Loss

As we have mentioned earlier, the adversarial loss is not enough to learn G and F . A generator only needs to ensure that the distribution of its outputs looks like the target domain, it does not have to map each input image to a meaningful output. Thus, a generator could output any image from the target domain, even random ones, and still fool the discriminator. This is what we previously described as *mode collapse*. To mitigate this and ensure meaningful translations, we introduce the cycle-consistency loss $\mathcal{L}_{cyc}$, which encourages the generator to preserve the content of the input while only translating style, defined as follows

$$\mathcal{L}_{cyc}(G, F) = \mathbb{E}_{x \sim \mathcal{D}_x} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim \mathcal{D}_y} [\|G(F(y)) - y\|_1].$$

The first term is forward cycle consistency Figure 3(b) and the second is backward cycle consistency Figure 3(c).

Full objective function

Using the above, we construct the full objective function

$$\begin{aligned} \mathcal{L}(G, F, D_X, D_Y) &= \mathcal{L}_{ADV}(G, D_Y, X, Y) \\ &\quad + \mathcal{L}_{ADV}(F, D_X, Y, X) \\ &\quad + \lambda \mathcal{L}_{cyc}(G, F) \end{aligned}$$

where lambda is a hyperparameter controlling the importance of cycle consistency in the total loss.

3.2.1 Grad-CAM

We expect our classifier learns to look at tissue structures and identify what is actually tumor tissue when it makes such predictions, and we will use Grad-CAM [Selvaraju et al. 2019] to check this.

Grad-CAM provides visual explainability by highlighting which regions of the input image had the most impact for the prediction given (ibid.). It uses the gradient information flowing from the last convolutional layer of the CNN to highlight what part of the input is important for the prediction. This is done by computing channel-wise *importance weights* by global average pooling the gradients. These weights and the feature maps are combined into the heatmap we are interested in and upsampled such that it can be overlayed on the input image. Another method is Grad-CAM++ [Chattopadhyay et al. 2018], that can provide better visual explanations if there are multiple object instances and provide more precise localization. One limitation of Grad-CAM is the low resolution of the feature maps. If a higher resolution is needed, one could use Guided Grad-CAM [Selvaraju et al. 2019], but we did not find a need for it in this project. Using Grad-CAM we can also verify that it does not look at scanner artifacts for cues.

To make this assessment clinically meaningful, we additionally overlay the tumor polygons extracted from the JSON annotation files onto the original patches.

3.3 Implementation Details

Throughout the entire project, we use the *PyTorch* framework. In our final version, the generator in the CycleGAN is a simple CNN with 3 downsampling blocks consisting of 2d convolutions, batch normalization and leaky ReLU. The batch normalization is added to follow the recommendation from lecture 7: ConvNet Architectures. Afterwards, it has 3 upsampling blocks consisting of transpose 2d convolutions, batch normalization and ReLU. It also has skip-connections as in a U-Net, we explain this thoroughly in the next section. The generator ends with tanh to match the normalization of the data in the range $[-1, 1]$.

The discriminator is a PatchGAN [Demir and Unal, 2018] with 14×14 patches. It consists of four blocks of 2d convolutions, batch normalization and leaky ReLU.

The classifier is also a CNN, but at the end it has fully connected layers to the number of classes, i.e. 2. It is a custom CNN using 3 blocks of 2d convolutions, max pooling, and ReLU before the fully connected layers. It was trained using the Adam optimizer [Kingma and Ba 2017] with a learning rate of $1 \cdot 10^{-5}$. In Appendix 6 we argue for that choice. It uses cross entropy as the loss function and was trained for 120 epochs.

We also did transfer learning, see Appendix [11](#). This was mostly such that we could compare our classifier to a “good” model.

In our implementation of Grad-CAM++ we use Grad-CAM with positive gradients as noted in [Lerma and Lucas \(2022\)](#).

3.4 Experiments

In this section, we cover some of our considerations when making our models and evaluate these considerations through some experiments.

Firstly, we describe our baseline model. The generator can be seen as an adversarial autoencoder in which its bottleneck layer is trained to match a target domain. Thus, our model consists of an encoder and a decoder which initially simply uses convolutions and ReLU activations with three encoder blocks and three decoder blocks. The discriminator a PatchGAN-style discriminator, a CNN with 5 blocks of convolutions using ReLU. By pure visual inspection of what the model generates (See figure [6](#)) it is obvious that the model needs some work.

Machine learning models sometimes suffer from *internal covariate shift*, and since it is a known fact that models benefit from batch normalization, we might as well add it to our models.

Another consideration we had was the number of zero activations in our model. Running a simple python script that counts the number of logits that output zero, we get a results showing that approximately half of them are zero. This can be a problem when nodes containing fine structural information are suppressed, likely hurting the gradient flow. To address this we replaced our ReLU activations with LeakyReLU(0.2), which will maintain a small gradient for negative inputs.

Typically encoder/decoder -like models tend to lose some information when approaching the bottleneck layer. Thus we immediately thought to use a U-Net inspired model, in which we add skip-connections to our generator model, which can help carry some of the fine-grained details through the network. [Krikid et al. \(2024\)](#) also found that “U-Net demonstrated superior performance in preserving fine-grained details crucial for microscopic imagery”. This aligns with the results of [Zhu et al. \(2017b\)](#) who used U-Net architectures to help improve the performance of CycleGANs. See figure [6](#) for the generation results, which shows much better results.

Although a visual inspection provides a qualitative assessment, a systematic quantitative evaluation is necessary to evaluate the models. It is known to be hard to evaluate GAN models. Since our model works on unpaired data, popular methods such as SSIM will not work. Instead, we can look at the cycle consistency, which is a good indication of performance. the idea being high cycle consistency implies that little information is lost.

Though cycle consistency reflects how much information is preserved, it does not directly measure how realistic or domain-aligned the generated images are. As an addition, we can use *Fréchet Inception Distance* (FID) [Heusel et al. \(2017\)](#), which is a standard method for evaluating GANs. FID compares the distributions of real and generated images using a pretrained network, measuring both the differences in means and covariances. Lower FID values corresponds to generated images that are closer to the real images. Since FID does not require paired data, it is a well suited for our setting.

3.4.1 Without CycleGAN

As part of the pipeline, we investigate how well the classifier performs on data from the other domain. We train the classifier on data from domain B , then test it on data from domain A and finally test it on adapted data from domain A using CycleGAN to see if it is an improvement.

3.4.2 Artifacts from CycleGAN

The CycleGAN may introduce artifacts when converting between domains. Therefore, we conduct an experiment where we check if the Grad-CAM is consistent in X and $G(F(X))$. If artifacts are introduced, then the model will look elsewhere than in the original image and may change its prediction.

4 Results

The classifier has a test accuracy of 98.63% when run on data from the domain it trained on. Compare this to Figure [11](#) in the appendix, where we used transfer learning to get an accuracy of 86%. Using the test data from the other domain we got an accuracy of 50.47% on our normal classifier. Its confusion matrix shows that it only predicts tumor.

Applying our trained GAN model, and then testing the classifier again shows a test accuracy of 80.6%, which is a clear improvement over the baseline, and very close to the performance achieved on the original data.

Our results align with what we have seen in the literature (Shaban et al. [2018], Runz et al. [2021], Vahadane et al. [2016]), and while we cannot directly compare our results to theirs due to differences in datasets and evaluation methods, we do see a similar pattern. The improvement gained from using a CycleGAN is high compared to non-normalized classification. Even with limited computing resources, we managed to achieve high accuracy on classification and great generated images (both from visual inspection and FID scores).

Figure 4 shows the Grad-CAM(++) heatmap for the 3 predictions with respectively worst and best loss. It shows that the model actually looks at the tissue and not the white background.

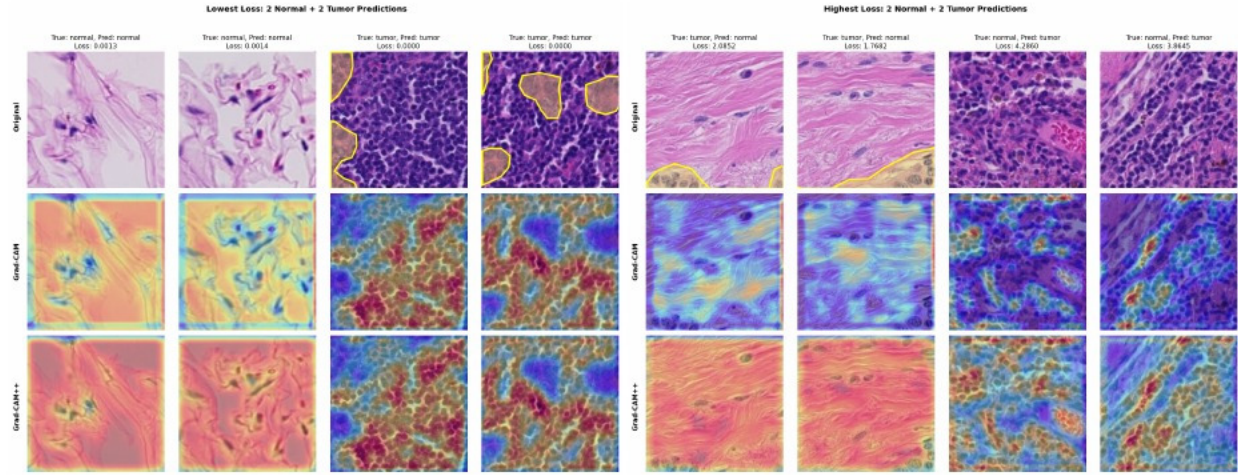


Figure 4: Top four images with best (lowest) and worst (highest) loss with two images from each class and their respective Grad-CAM(++) heatmap.

	Predicted Normal	Predicted Tumor
Actual Normal	260	8
Actual Tumor	99	164

Table 1: Confusion matrix for our classifier on test data.

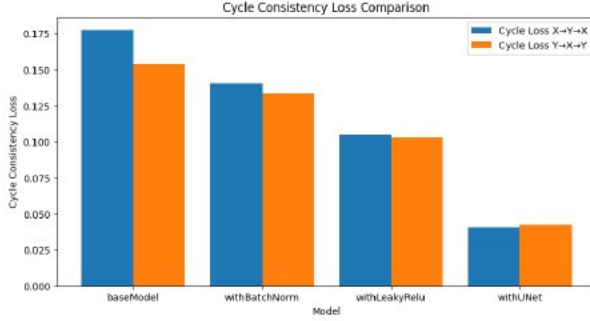
Comparing all the models we implemented, it is evident that the U-Net version was the best. Looking at Figure 5a we see that each model gets slightly better and that the relative difference is noticeable. However, we do need to remember that the model is trained to minimize this loss, and what we really care about is its ability to translate between domains. This is where FID comes in. We see that the addition of LeakyReLU does not change the score much. In fact, it is slightly worse. The takeaway from these results is that the U-Net version was definitely better than the baseline, which is consistent with the results from Zhu et al. [2017b]. This is also clear by visual inspection (see Figure 6), where it is also unclear if the LeakyReLU version is even better than the BatchNorm version.

All GAN models were trained in 50 epochs (see loss curves in Figure 9 in the Appendix). From our loss curves, we think that they could have trained a little longer, but overall it seems to have converged nicely.

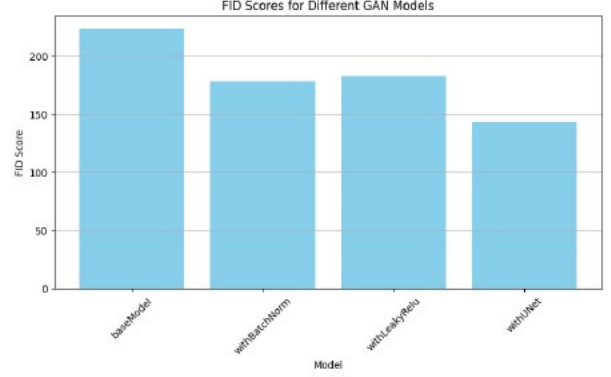
5 Discussion

In Table 1 we see that our model is neutral and has the highest value along the diagonal. This suggests that our model is not guessing and has learned to recognize tumors.

Although the results indicate that our model is able to distinguish between tumors and normal lesion patches with high accuracy, it might not be appropriate in a medical context. In medical applications, the priority would be to minimize false negatives (Actual Tumor / predicted as normal), since misclassifying a patient carrying a tumor carries serious consequences for the patient. While false positives are comparatively benign as they would primarily result in additional inspections. We could change the threshold such that the model predicts normal only when it is highly confident.



(a) Cycle consistency comparison of the different models.



(b) FID comparison of the different models.

Figure 5

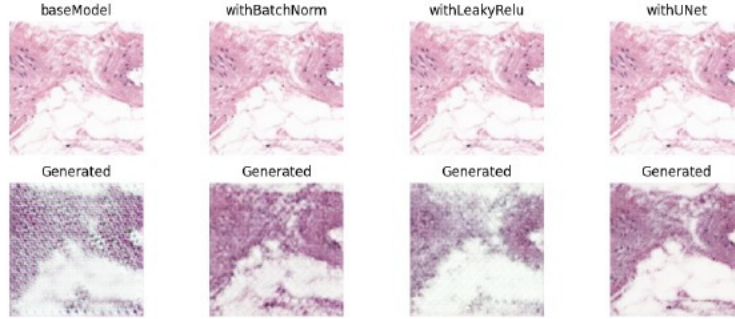


Figure 6: Generated images from all our models

However, adjusting the threshold would only affect the final prediction. It does not guarantee that the model is basing its decisions on anything meaningful.

As revealed by the Grad-CAM and Grad-CAM++ heatmaps in Figure 4 while the model predicts with high confidence, the attention is specifically on the normal tissue instead of the tumor tissue which is most clear in the heatmaps for the lowest loss where the true label is tumor. It therefore shows that the model is learning superficial texture cues rather than clinically relevant features, meaning that improving the threshold would reduce false negatives numerically, but would not resolve the underlying issue.

Part of this behavior is explained by how our dataset is constructed. We use the default extraction settings, i.e. a minimum tissue fraction of 0.5 and a minimum tumor fraction of 0.02 for tumor patches. This removes completely empty or almost-empty patches, but still allows many tumor patches where only a small part of the 128×128 crop contains tumor, often near the border, while most of the patch looks normal. In such cases the model can achieve a correct tumor label by relying on background texture or staining cues rather than on clear malignant structures, which is exactly what our Grad-CAM overlays suggest.

However, we chose to use these thresholds because they provide more training patches. Increasing the minimum tumor fraction would reduce the noise in the labels, but it would also eliminate a large portion of the tumor dataset, which is already much smaller than the normal class. Since our primary goal was to study the effect of stain normalization rather than to build a clinically optimized classifier, retaining more patches was preferable even if it introduced some label noise.

5.1 Artifacts from CycleGAN

The goal of this experiment is to verify if transforming an image from one domain to another and back preserves the clinical semantics. We compute the Pearson product-moment correlation coefficients (`np.corrcoef(...)`) of the original attention maps and the reconstructed attention maps. Figure 7a shows an ideal outcome, where the reconstruction is perfect to the human eye and the classifier “sees” the same thing in both images. The attention maps

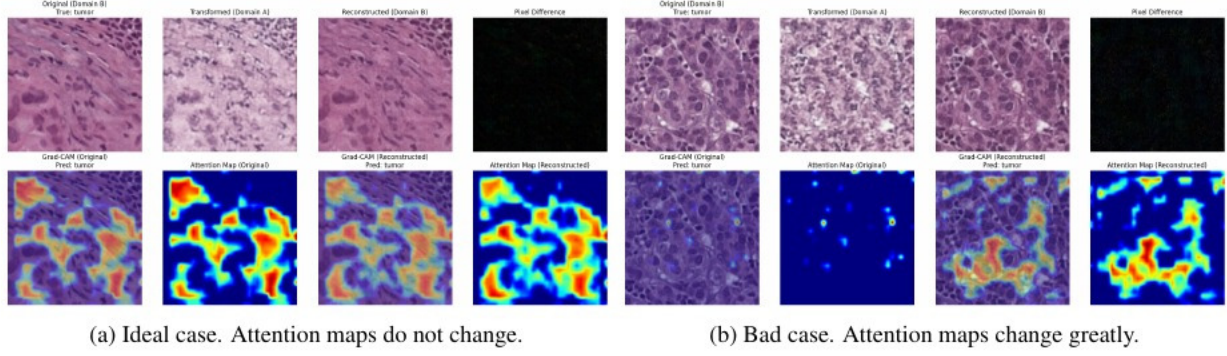


Figure 7

match almost perfectly, meaning the transformation process did not hallucinate new features or erase existing clinical markers that affect the classifier. It has a correlation of 0.9188.

Figure 7b shows a failure case. Even though to the reconstruction seems identical to the human eye, there is a massive discrepancy in model attention. The prediction is still the same, but it is worrisome that the attention has changed so much. It has a correlation of 0.1259.

Figure 8 shows the consistency distribution is heavily skewed toward the right with a massive peak near 1.0, indicating that the vast majority of image reconstructions preserve the model’s attention perfectly, resulting in an overall mean correlation of 0.740 as marked by the red dashed line.

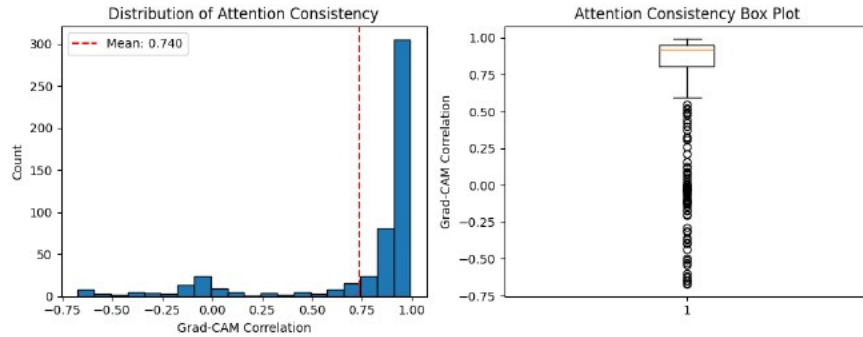


Figure 8

Our results demonstrate that while domain adaptation significantly improves classification, high visual fidelity in reconstruction does not guarantee the preservation of diagnostic features. Using attention-based consistency metrics are an essential addition to the validation pipeline to detect semantic discrepancies that standard performance metrics miss that can cause issues down the line.

6 Future work

For future work, it would be interesting to explore stain normalization across more than two centers. [Shaban et al., 2018, Runz et al., 2021] show that using data from several hospitals can make models more robust to different staining styles. Extending our setup to multiple centers would therefore make the task more realistic and allow us to test whether CycleGAN can generalize to a wider range of staining variations.

Another idea is to make a model that actually bases its predictions on the actual tumor tissue, which we saw using Grad-CAM that it did not, and whether this model would be better.

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Appendix

Loss-curves for CycleGAN

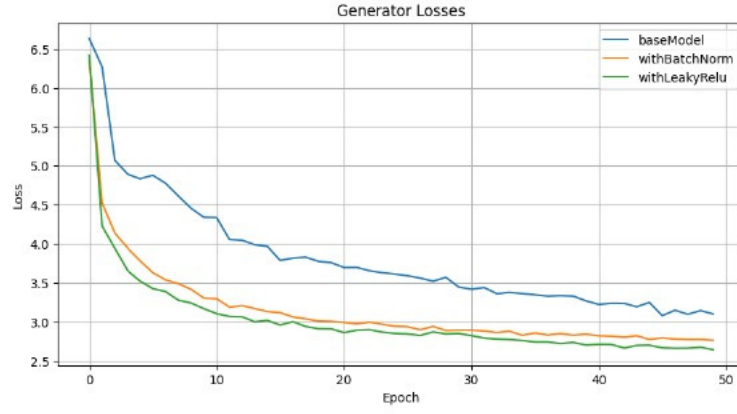


Figure 9: The generator loss curves of training all our CycleGAN models. Loss curves suggest the models had not fully converged; however, training was terminated at 50 epochs due to computational constraints.

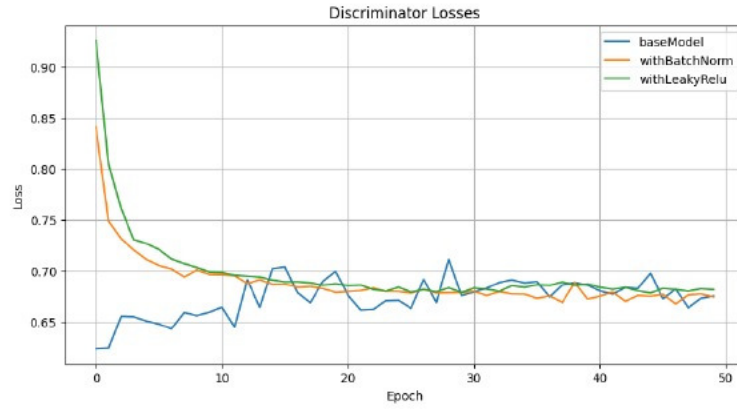


Figure 10: The discriminator loss curves of training all our CycleGAN models

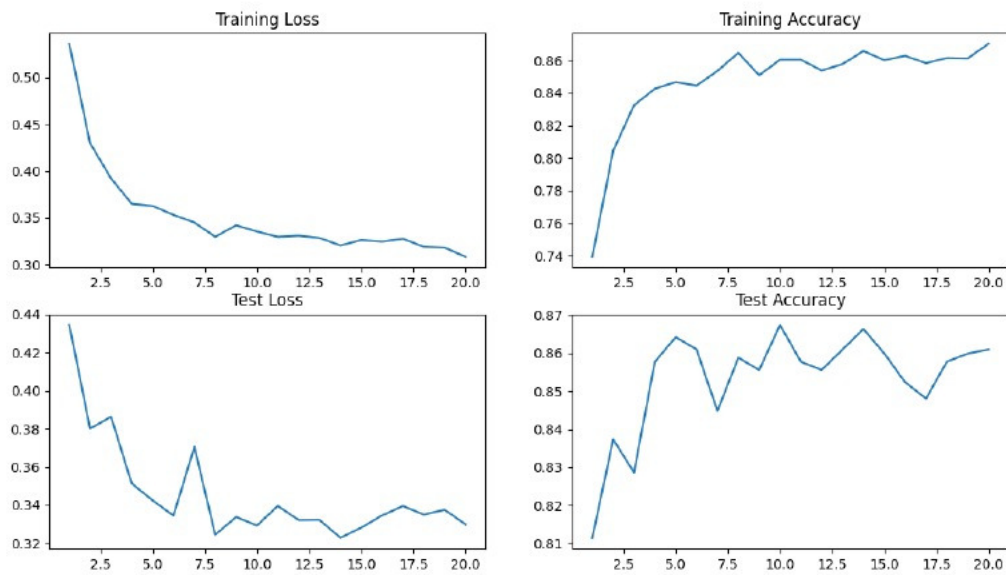


Figure 11: Loss/Accuracy when doing transfer learning on one of the domains for the classifier.

Hyperparameters for Classifier

Figure 12 shows the loss curves and accuracy of the classifier trained with different learning rates and optimizers. Therefore, we set the learning rate to $1e-5$ and used Adam to train the classifier, but it had not finished learning in 100

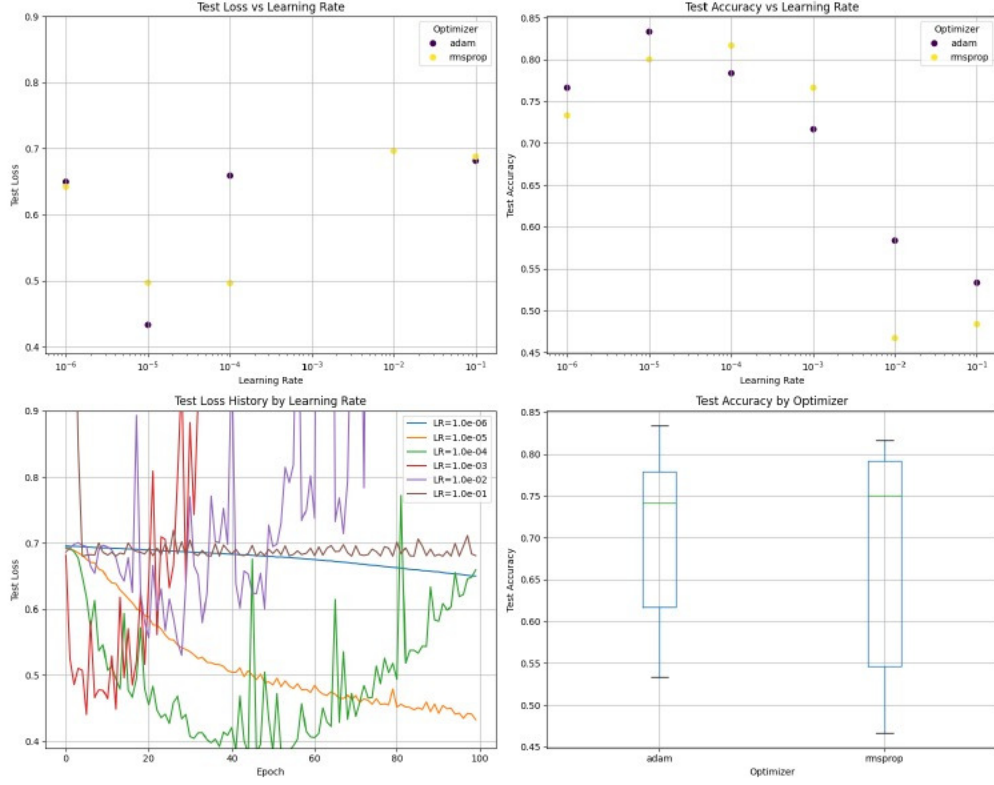


Figure 12: Trying the combinations of $(LR, \text{optimizer}) \in \{1e^{-6}, 1e^{-5} \dots 1e^{-1}\} \times \{\text{Adam}, \text{RMS}\}$.

epochs and should train longer. Figure 13 shows the loss curves for the used classifier.

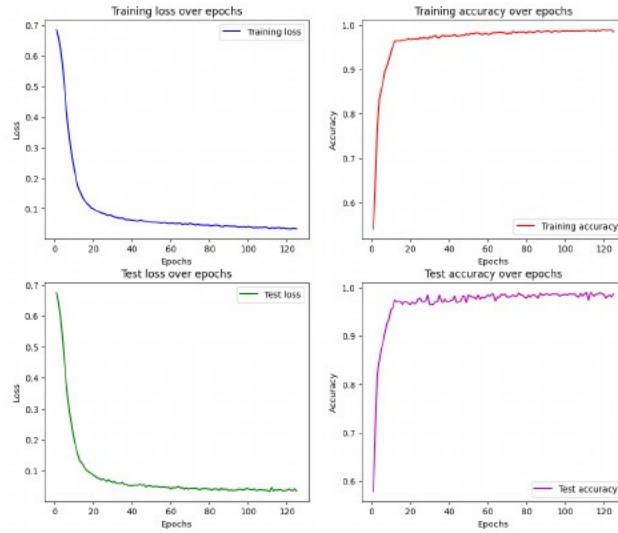


Figure 13

GAI Declaration

We used ChatGPT-5 (3/12-2025) and Gemini 3 Pro (4/12-2025) to criticize our report and provide feedback. We used the following prompt

Give suggestions on what can be done to improve the report and give critical feedback, but do not actually implement them or write it for me. Only provide helpful questions and constructive points of criticism that can be used to improve the report.

It is inspired by the prompt Niels Olof presented on usage of GAI on the 12th of March 2025.